Genetics and population analysis

PhyloAE: Finding Admixed Dog Breeds from Embedding Geometry

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Abstract

Motivation: We want a 2D or 3D scatter of the Parker et al. (2017) canine SNP panel (1,337 dogs) where each breed sits in its assigned Parker clade. UMAP, t-SNE, PaCMAP, TriMAP, and PHATE optimize one geometric scale and do not preserve that breed-within-clade hierarchy, so admixed breeds get placed in the wrong clade and disappear into neighboring clusters.
Results: PhyloAE adds two MDS-stress terms to a standard autoencoder loss, one on breed centroids and one on clade centroids, with targets fixed in PCA space. On the Parker et al. (2017) panel (1,337 dogs, 175 breeds, 23 clades), PhyloAE reaches 64.0% strict Parker concordance against 50.9% for the best baseline, 1.56× ADMIXTURE ARI at $K{=}8$, and Mantel $r{=}0.588$ against Parker's NJ tree (1.20× t-SNE). All six methods agree on misplacing 17 breeds; 11 fall inside the groups Parker et al. flagged as ambiguous (U.K. Rural and Mediterranean), 6 are new candidates.
Availability and implementation: Source code: github.com/aariya50/phyloae-dog-breeds. Python 3.11, MIT license. Data: NHGRI Parker 2017 panel.
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Supplementary information: Figures S1 (cluster estimation), S2 (NJ-tree bootstrap), and S3 (admixture-score distribution) are committed to the repository under `results/figures/`.

1 Introduction

The Parker et al. (2017) SNP panel groups 1,337 dogs into 175 breeds and 23 clades. The standard low-D embeddings of this panel optimize a single geometric objective: UMAP and PaCMAP tighten neighborhoods, t-SNE and TriMAP focus on local distances, PHATE smooths diffusion trajectories. None of them puts the breed-within-clade hierarchy into the loss, so breeds that sit between clades land somewhere reasonable for the local objective but not in a way that respects the Parker et al. NJ tree.

PhyloAE is an autoencoder with two extra MDS-stress terms in the training loss: one on the 175 breed centroids, one on the 23 Parker clade centroids. The targets are pairwise distances computed in PCA space once and frozen. The trained model places breeds so they preserve the breed-within-clade hierarchy, and the breeds where the embedding consistently disagrees with the assigned clade turn out to fall in groups Parker et al. (2017) themselves called ambiguous, especially U.K. Rural sighthounds and herding dogs, and Mediterranean livestock guards.

2 Methods

2.1 Data

The analysis uses the Parker et al. (2017) genotype panel. After quality control, LD pruning, and autosome filtering, we're left with 1,337 purebred dogs and 63,285 SNPs across 175 breeds and 23 canonical clades, plus a Basenji singleton, one unassigned breed, and 6 wolves as outgroup (26 labels total).

2.2 Baseline embeddings

Five unsupervised baselines, UMAP, PaCMAP, t-SNE, TriMAP, and PHATE, receive the same scaled SNP matrix with no breed or clade labels, each evaluated through a hyperparameter sweep. A factorial sweep covers 11,770 configurations across UMAP, PaCMAP, t-SNE, TriMAP, PHATE, and PhyloAE (5 seeds, 2D + 3D). Per-method sweep grids and best-config selection are reproducible from `phyloae.ipynb` in the project repository.

2.3 PhyloAE: hierarchical MDS autoencoder

PhyloAE is a symmetric autoencoder with a low-dimensional bottleneck (evaluated at 2D and 3D; headline uses 3D), $63285 \to 512 \to 64 \to k\mathrm{D} \to 64 \to 512 \to 63285$ with $k \in \{2, 3\}$. Hidden layers use batch normalization and ReLU. Training minimizes reconstruction error plus two centroid-distance stress terms,

$$L = L_{\mathrm{recon}} + \lambda_b\, L_{\mathrm{breed}} + \lambda_c\, L_{\mathrm{clade}}. \tag{1}$$

Centroid stress at each scale is normalized MDS stress over all centroid pairs,

$$L_{\mathrm{scale}} = \frac{\sum_{i<j} \left(d_{\mathrm{embed}}(c_i, c_j) - d_{\mathrm{target}}(c_i, c_j)\right)^2}{\sum_{i<j} d_{\mathrm{target}}(c_i, c_j)^2}. \tag{2}$$

Targets are fixed pairwise distances between centroids in 20-dimensional PCA space, max-normalized to $[0, 1]$. L_{breed} uses 175 breed centroids; L_{clade} uses 26 label centroids (23 Parker clades plus a Basenji singleton, an unassigned breed, and the wolf outgroup). Labels aggregate encoded samples into centroids for loss computation only, never as encoder input features. Training uses Adam, full-batch, 500

Table 1. Best configuration per method.

Method	Dim	Trust.	Silh.	Comp.	Seed
t-SNE	2D	**0.932**	0.841	**1.773**	0.86
PhyloAE	3D	0.915	0.774	1.689	**0.61**
PaCMAP	3D	0.833	**0.856**	1.688	0.51
UMAP	3D	0.870	0.818	1.688	**1.00**
PHATE	3D	0.897	0.627	1.524	0.63
TriMAP	3D	0.776	0.069	0.845	0.70

Trust. = trustworthiness; Silh. = silhouette; Comp. = Trust. + Silh.; Seed = seed-stability.

epochs. Headline configuration: `hidden_dims`=(512, 64), `lr`=3e-4, $\lambda_b = \lambda_c = 0.1$, `target_metric`=cosine, $n_{\text{components}} = 3$, over five random seeds (seed 789 used for the figures). The cosine-3D configuration was the dimension-agnostic Pareto winner of a 720-run sweep over the three candidate target metrics (euclidean, manhattan, cosine).

2.4 Evaluation metrics

We score each method on local-neighborhood fidelity (trustworthiness), cluster separation (silhouette over breed labels), and reproducibility across five random seeds (seed stability). Each method is evaluated at 2D and 3D; the best composite (trustworthiness + silhouette) is selected per method. All downstream analyses use each method's best-composite configuration: t-SNE at 2D, all other methods at 3D. External validation uses ADMIXTURE-derived dominant-component labels across K=2..30, with ARI and NMI. Parker concordance: a breed is correct only if its nearest clade centroid in the embedding is its Parker-assigned clade.

3 Results

3.1 Best-configuration selection

Table 1 reports each method's best configuration at its best-composite dimension. t-SNE leads on trustworthiness; PaCMAP on silhouette.

3.2 Parker concordance

Figure 1 reports strict Parker concordance per method at the best-composite dimension. PhyloAE leads at 64.0%; the unsupervised baselines fall between 30.3% and 50.9%. The same ordering recurs in ADMIXTURE agreement.

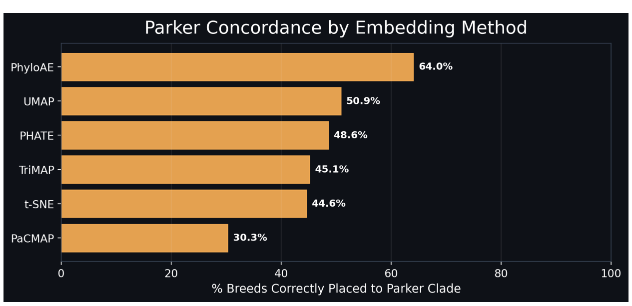

Fig. 1. Parker concordance across methods at best-composite dimension.

Seventeen breeds are misplaced by all six methods (17/17 consensus). They break down into three groups. Eleven sit in clades Parker et al. (2017) flagged as ambiguous in their own discussion (see Table 2): seven U.K. Rural breeds (4 sighthounds plus 3 herders) and four Mediterranean breeds (2 ancient sighthounds plus 2 livestock guards). The remaining six are not anticipated by that text (see Table 3): three small terriers in the Terrier clade plus three singletons (Great Dane, Brussels Griffon, Portuguese Water Dog) each sitting outside their assigned clade.

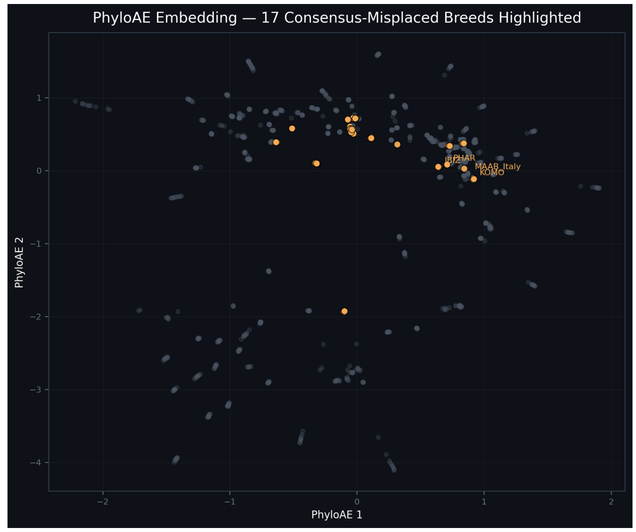

Fig. 2. The 17 consensus-misplaced breeds (orange) on the PhyloAE embedding (first two components of the 3D bottleneck, cosine target). Eleven anticipated in Parker et al. (2017) (U.K. Rural + Mediterranean) and six novel candidates (three terriers in the Terrier clade plus three singletons across distinct clades).

Table 2. Eleven anticipated, "both the U.K. Rural and the Mediterranean clades include both sighthound and working dog breeds" (Parker et al., 2017).

Code	Breed	Parker clade	Sub-type
BORZ	Borzoi	UKRural	sighthound
GREY	Greyhound	UKRural	sighthound
ITGY	Italian Greyhound	UKRural	sighthound
IWOF	Irish Wolfhound	UKRural	sighthound
AUSS	Australian Shepherd	UKRural	herder
BORD	Border Collie	UKRural	herder
KELP	Kelpie	UKRural	herder
IBIZ	Ibizan Hound	Mediterranean	ancient sighthound
PHAR	Pharaoh Hound	Mediterranean	ancient sighthound
KOMO	Komondor	Mediterranean	livestock guard
MAAB	Maremma Sheepdog	Mediterranean	livestock guard

Table 3. Six novel candidates.

Code	Breed	Parker clade
AIRT	Airedale Terrier	Terrier
SILK	Silky Terrier	Terrier
YORK	Yorkshire Terrier	Terrier
DANE	Great Dane	EuropeanMastiff
BRUS	Brussels Griffon	ToySpitz
PTWD	Portuguese Water Dog	Poodle

3.3 ADMIXTURE concordance

ADMIXTURE provides an independent allele-frequency reference without embeddings or labels. Concordance peaks for most methods around $K{=}8$ (Figure 3); we report that operating point. At $K{=}8$, PHYLOAE achieves ARI$=$0.149 versus PHATE 0.095 (1.56$\times$). Across all 29 K values, PHYLOAE leads at the majority of K values. KMeans silhouette curves (Supp. Fig. S1) show all methods peak above $K{=}23$, the canonical Parker partition is useful but coarse. A neighbor-joining tree on PHYLOAE breed centroids shows 127/172 internal splits with bootstrap $> 70\%$ (Supp. Fig. S2), confirming hierarchical signal.

Fig. 3. ADMIXTURE concordance versus K across methods.

3.4 Extended validation

Mantel correlation. Pairwise breed-centroid distances in each embedding versus Parker et al. (2017)'s original NJ tree (supplement mmc5), collapsed to breed-mean patristic distances across 175 breeds in our panel, evaluated with the Mantel test: PHYLOAE achieves $r{=}0.588$, 1.20$\times$ the next best (t-SNE, 0.490; Figure 4).

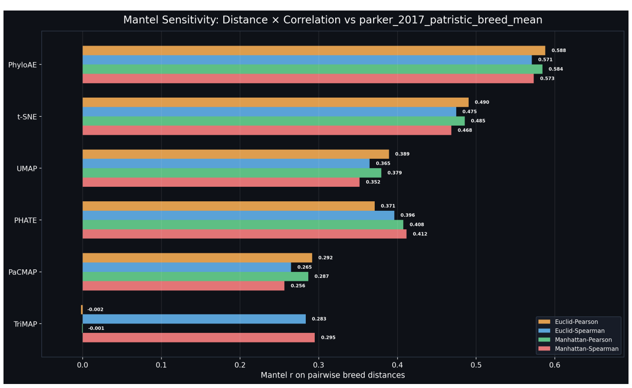

Fig. 4. Mantel correlation versus Parker et al. (2017) NJ-tree distances.

Consensus admixture score. For each breed we average the ratio $d_{\text{own}}/d_{\text{nearest}}$ (own-clade-centroid distance over nearest-other-clade distance) across all 6 methods. The 10 breeds with the highest score are all Parker-labeled Mediterranean (Supp. Fig. S3), reinforcing the consensus-misplaced finding.

4 Discussion

PHYLOAE sees breed and clade labels at training time, so the obvious worry is that it is just memorizing the labels. Three checks push back. KMeans silhouette over the embedding peaks at $K \gg 23$ rather than at $K{=}23$, so the model isn't snapping to the label count. ADMIXTURE only sees allele frequencies and still agrees with PHYLOAE's geometry at $K{=}8$ (ARI 0.149, 1.56$\times$ PHATE), and that agreement can't come from Parker et al. (2017). The Mantel correlation against Parker's continuous NJ-tree distances ($r{=}0.588$, 1.20$\times$ t-SNE) tests at finer resolution than the 23-bin projection.

The two-scale supervision is the only thing different from a plain MDS autoencoder. One stress term trades coarse against fine; using both at once keeps both. PHYLOAE doesn't take the trustworthiness lead in Table 1 (t-SNE does), but it does have the strongest external concordance.

The 17-breed consensus disagreement is the actual result. Every one of those breeds is placed wrong by all six methods, so the disagreement isn't a quirk of any one method. The breeds themselves cluster in groups that are already known to be borderline: U.K. Rural sighthounds and herding dogs, Mediterranean livestock guards and ancient sighthounds, and small terriers. A few caveats: the 23 Parker et al. clades are a hand-picked simplification (cluster-estimation favors a finer K), PHYLOAE's supervised loss limits strictly unsupervised comparability (the label-free ADMIXTURE check is the mitigation), and strict concordance is harsh on near-misses, but none of them dissolves the cross-test agreement on the 17 breeds.

5 Conclusion

Two MDS-stress terms on top of a vanilla autoencoder are enough to recover the breed-within-clade hierarchy on the Parker 2017 panel. PHYLOAE matches t-SNE on trustworthiness (0.915 vs. 0.932) and clears the unsupervised baselines on every external check (Parker concordance, K-wise ADMIXTURE agreement, Mantel correlation against the Parker et al. NJ tree). The real payoff is the 17-breed cross-method consensus: 11 breeds line up with groups Parker et al. (2017) called borderline, and 6 are novel candidates for follow-up genotyping (three small terriers in the Terrier clade plus Great Dane, Brussels Griffon, and Portuguese Water Dog).

The two-scale loss should work on any dataset with both coarse and fine labels (here, breed = fine, clade = coarse). The PCA targets used here are the simplest choice; if a curated tree or other reference distance is available, it can replace the PCA targets without changing the rest of the model. We haven't tested this on non-canine panels.

Availability and Implementation

The full pipeline lives in a single Jupyter notebook, `phyloae.ipynb`, with importable utilities under `helpers/`. The only piece written from scratch is the PHYLOAE model and its training loop (`helpers/lib.py`). Everything else uses standard packages: PyTorch for the autoencoder, scikit-learn for t-SNE / PCA / metrics (trustworthiness, silhouette, ARI, NMI), UMAP-learn, PaCMAP, TriMAP, and PHATE for the unsupervised baselines, scikit-bio for the neighbor-joining tree, PLINK 1.9 for QC and LD pruning, and ADMIXTURE 1.3 for ancestry estimation. Source code: github.com/aariya50/phyloae-dog-breeds (MIT licensed). Data: NHGRI Parker 2017 SNP panel; a 50-dog subsample reproduces the pipeline end-to-end in roughly five minutes.

References

Parker, H. G., Dreger, D. L., Rimbault, M., Davis, B. W., Mullen, A. B., Carpintero-Ramirez, G., and Ostrander, E. A. (2017). Genomic analyses reveal the influence of geographic origin, migration, and hybridization on modern dog breed development. *Cell Reports*, **19**(4), 697–708.